# Discrete and Continuous Markov Decision Processes: Theory, Algorithms, and Mathematical Illustrations

**Abstract**

This report presents a mathematical exposition of Markov Decision Processes (MDPs), extending from discrete dynamic programming methods to continuous-state control and simulation-based approximation. Detailed worked examples accompany value iteration, policy iteration, finite-horizon backward induction, Monte Carlo estimation, fitted value iteration, and Linear Quadratic Regulation (LQR). The goal is to bridge reinforcement learning and optimal control through explicit derivations and numerical demonstrations.

## 1 Discrete Markov Decision Processes

An MDP is defined as $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$.

The Bellman optimality equation is:

$$V^*(s) = \max_a \left[ R(s,a) + \gamma \sum_{s'} P(s'|s,a) V^*(s') \right]$$

### 1.1 Example 1: Value Iteration (Two-State MDP)

Let:

$$\mathcal{S} = \{s_1, s_2\}, \quad \mathcal{A} = \{a_1, a_2\}$$

Transitions:
If in $s_1$: - $a_1$: stays in $s_1$ with prob 1, reward = 2 - $a_2$: goes to $s_2$ with prob 1, reward = 0
If in $s_2$: - any action: stays in $s_2$, reward = 1
Let $\gamma = 0.9$.
Initialize $V_0(s_1) = V_0(s_2) = 0$.
Iteration 1:

$$V_1(s_1) = \max\{2 + 0.9(0),\ 0 + 0.9(0)\} = 2$$

$$V_1(s_2) = 1 + 0.9(0) = 1$$

Iteration 2:

$$V_2(s_1) = \max\{2 + 0.9(2),\ 0 + 0.9(1)\} = \max\{3.8, 0.9\} = 3.8$$

$$V_2(s_2) = 1 + 0.9(1) = 1.9$$

Continuing iterations yields convergence toward:

$$V^*(s_1) = \frac{2}{1-0.9} = 20, \quad V^*(s_2) = \frac{1}{1-0.9} = 10$$

Thus optimal action in $s_1$ is always $a_1$.

## 2 Policy Iteration Example

Suppose we start with policy $\pi(s_1) = a_2$, $\pi(s_2) = a_1$.

Policy evaluation solves:

$$V(s_1) = 0 + 0.9V(s_2)$$
$$V(s_2) = 1 + 0.9V(s_2)$$

Solve second equation:

$$V(s_2) = \frac{1}{1-0.9} = 10$$

Then:

$$V(s_1) = 0.9(10) = 9$$

Policy improvement:

$$Q(s_1, a_1) = 2 + 0.9(9) = 10.1$$

$$Q(s_1, a_2) = 0 + 0.9(10) = 9$$

Thus policy updates to $a_1$ in $s_1$.

## 3 Finite Horizon Example

Horizon $T = 2$.

Terminal value:

$$V_2(s) = 0$$

Backward step:

$$V_1(s_2) = 1$$

$$V_1(s_1) = 2$$

At $t = 0$:

$$V_0(s_1) = \max\{2 + 0.9(2),\ 0 + 0.9(1)\} = 3.8$$

Finite-horizon values differ from infinite-horizon fixed points.

## 4 Continuous-State MDP

Consider:

$$s_{t+1} = s_t + a_t + \epsilon_t$$
$$\epsilon_t \sim \mathcal{N}(0, \sigma^2)$$
$$R(s, a) = -s^2 - a^2$$

Bellman equation:

$$V(s) = \max_a \left[-s^2 - a^2 + \gamma \mathbb{E}_\epsilon[V(s + a + \epsilon)]\right]$$

If we approximate:

$$V(s) = -ks^2$$

Then:

$$\mathbb{E}[V(s + a + \epsilon)] = -k\mathbb{E}[(s + a + \epsilon)^2]$$

$$= -k[(s + a)^2 + \sigma^2]$$

Substitute into Bellman equation:

$$V(s) = \max_a \left[-s^2 - a^2 - \gamma k((s + a)^2 + \sigma^2)\right]$$

Optimization yields linear feedback $a = -\alpha s$.

## 5 Monte Carlo Approximation Example

Suppose we sample:

$$s'_1 = 1.2, \quad s'_2 = 0.8, \quad s'_3 = 1.1$$

If $V(s) = s^2$ then:

$$\mathbb{E}[V(s')] \approx \frac{1}{3}(1.44 + 0.64 + 1.21) = \frac{3.29}{3} = 1.0967$$

Thus Bellman backup uses empirical expectation.

## 6 Fitted Value Iteration Example

Suppose:

$$\hat{V}(s) = \theta_0 + \theta_1 s$$

Data:

$$(s_i, y_i) = (0, 1), (1, 2), (2, 5)$$

Regression minimizes:

$$\sum_i (\theta_0 + \theta_1 s_i - y_i)^2$$

Normal equations yield:

$$\theta_1 = 2, \quad \theta_0 = 0$$

Thus approximation: $\hat{V}(s) = 2s$.

# 7 Linear Quadratic Regulator (Scalar Case)

Let:

$$s_{t+1} = s_t + a_t$$

$$J = \sum_{t=0}^{\infty} (s_t^2 + a_t^2)$$

Assume:

$$V(s) = Ps^2$$

Bellman equation:

$$Ps^2 = \min_a \left[ s^2 + a^2 + P(s+a)^2 \right]$$

Expand:

$$= s^2 + a^2 + P(s^2 + 2sa + a^2)$$

Differentiate w.r.t. $a$:

$$2a + 2P(s+a) = 0$$

$$a(1+P) = -Ps$$

$$a^* = -\frac{P}{1+P} s$$

Substitute back:

$$P = 1 + P - \frac{P^2}{1+P}$$

Multiply both sides by $(1+P)$:

$$P(1+P) = (1+P)^2 - P^2$$

Solving gives:

$$P = \frac{1+\sqrt{5}}{2}$$

Thus optimal feedback gain:

$$K = \frac{P}{1+P}$$

# 8 Nonlinear Linearization Example

Suppose:

$$s_{t+1} = s_t + a_t + s_t^2$$

Linearize around $s^* = 0$, $a^* = 0$:

$$A = \frac{\partial}{\partial s}(s + s^2)\bigg|_0 = 1$$

$$B = \frac{\partial}{\partial a}(a)\bigg|_0 = 1$$

Thus local linear model:

$$s_{t+1} \approx s_t + a_t$$

LQR can now be applied locally.

# 9 Theoretical Foundations and Additional Concepts

## 9.1 Action-Value Function (Q-Function)

In addition to the state-value function $V^\pi(s)$, it is often useful to define the action-value function:

$$Q^\pi(s,a) = R(s,a) + \gamma \sum_{s'} P(s'|s,a) V^\pi(s').$$

The Q-function represents the expected return of taking action $a$ in state $s$ and then following policy $\pi$ thereafter.

The optimal Q-function satisfies the Bellman optimality equation:

$$Q^*(s,a) = R(s,a) + \gamma \sum_{s'} P(s'|s,a) \max_{a'} Q^*(s',a').$$

The optimal value function is recovered as:

$$V^*(s) = \max_a Q^*(s,a).$$

Working with $Q$-functions is particularly useful when policies are derived directly via greedy maximization.

## 9.2 Deterministic and Stochastic Policies

A policy may be:

- **Deterministic:** $\pi : \mathcal{S} \to \mathcal{A}$
- **Stochastic:** $\pi(a|s)$ defines a probability distribution

In finite MDPs with discounted rewards, there always exists an optimal deterministic policy:

$$\pi^*(s) = \arg\max_a Q^*(s, a).$$

In continuous control problems such as LQR, optimal policies are typically deterministic and linear in the state.

## 9.3 Bellman Operator as a Contraction

Define the Bellman optimality operator:

$$(TV)(s) = \max_a \left[ R(s, a) + \gamma \sum_{s'} P(s'|s, a)V(s') \right].$$

**Theorem:** $T$ is a $\gamma$-contraction under the sup norm:

$$\|TV - TW\|_\infty \leq \gamma \|V - W\|_\infty.$$

**Proof Sketch:**

$$|(TV)(s) - (TW)(s)| \leq \gamma \max_a \sum_{s'} P(s'|s, a)|V(s') - W(s')|.$$

Since probabilities sum to 1,

$$\leq \gamma \|V - W\|_\infty.$$

Taking supremum over $s$ yields the result.
By the Banach fixed-point theorem, $T$ has a unique fixed point $V^*$ and value iteration converges:

$$V_{k+1} = TV_k \to V^*.$$

## 9.4 Continuous Action Optimization

In continuous control, actions belong to $\mathbb{R}^m$:

$$V(s) = \max_{a \in \mathbb{R}^m} \left[ R(s, a) + \gamma \mathbb{E}[V(f(s, a, w))] \right].$$

Thus solving the Bellman equation requires continuous optimization at every state. In special structured cases such as quadratic cost and linear dynamics, this optimization admits closed-form solutions (e.g., LQR).

### 9.5 Certainty Equivalence in Linear Quadratic Regulation

Consider stochastic linear dynamics:

$$s_{t+1} = As_t + Ba_t + w_t,$$

where $w_t$ is zero-mean noise.

A fundamental result is **certainty equivalence**: The optimal feedback gain $K$ derived from the deterministic system ($w_t = 0$) remains optimal in the stochastic case.

Noise affects the expected cost but does not alter the optimal linear control law:

$$a_t = -Ks_t.$$

This result highlights the structural robustness of LQR.

### 9.6 Approximation and Stability Considerations

When value functions are approximated:

$$V(s) \approx \hat{V}(s; \theta),$$

the Bellman operator is no longer guaranteed to be a contraction after projection onto the function class.

Errors compound through iterations:

$$\hat{V}_{k+1} = \Pi T \hat{V}_k,$$

where $\Pi$ denotes projection.

Unlike tabular value iteration, convergence is no longer guaranteed. This introduces important stability considerations in approximate dynamic programming and reinforcement learning.

### 9.7 Reinforcement Learning and Optimal Control: A Unifying View

Discrete MDPs, continuous-state control, simulation-based approximation, and LQR share a common mathematical backbone: the Bellman equation.

- Reinforcement learning emphasizes learning value functions from data.
- Optimal control emphasizes exploiting known dynamics for analytical solutions.
- Both solve fixed-point equations derived from dynamic programming principles.

The transition from tabular MDPs to continuous control represents not a conceptual shift, but an increase in dimensionality and structural complexity within the same foundational framework.

## 10 Conclusion

Through discrete dynamic programming, continuous-state Bellman equations, Monte Carlo simulation, regression-based approximation, and LQR optimal control, we see a unifying mathematical structure. Discrete MDPs provide conceptual clarity; continuous extensions require approximation; control theory supplies analytic solutions where structure permits.

Together, they form the backbone of modern reinforcement learning and optimal control.